

LLM TRAINING & SERVING

- LLM Fine-tuning
- LoRA / QLoRA
- Distributed Training
- LLM Evaluation
- OpenAI-Compatible Serving
- Multi-Task Adapters

MODELLING & ANALYTICAL METHODS

- Classical ML
- Deep Learning
- Computer Vision
- Natural Language Processing (NLP)
- Time-Series Forecasting
- Feature Engineering
- Behavioral Anomaly Detection
- Autoencoders
- Operations Research
- Vehicle Routing
- Mixed-Integer Programming
- Constraint Modelling
- Supply-Chain Optimization

FRAMEWORKS & APIS

- FastAPI
- scikit-learn
- XGBoost
- PyTorch
- TensorFlow / Keras
- TensorFlow Serving
- Hugging Face Transformers / PEFT
- Unsloth
- vLLM
- Google OR-Tools

GPU & DISTRIBUTED COMPUTING

- CUDA GPU Libraries
- GPU-Accelerated Computing
- RAPIDS cuDF / CuPy

EXPERIMENTATION & MODEL LIFECYCLE

- MLflow
- Experiment Tracking
- Model Evaluation
- Model Versioning / Registry
- Automated Retraining
- Vertex AI Pipelines
- Azure ML Pipelines
- Vertex AI Feature Store

CLOUD & SERVERLESS SERVICES

- GCP
- Microsoft Azure
- GKE
- Vertex AI
- Azure ML
- Cloud Run
- Cloud Functions
- Azure Function Apps
- Logic Apps

DEPLOYMENT & PRODUCTION RELIABILITY

- ML System Architecture
- Cloud-Native ML Systems
- Docker
- Kubernetes
- Helm
- Seldon Core
- Istio
- HPA Autoscaling
- CI/CD
- Cloud Build
- Azure DevOps / Pipelines
- ArgoCD
- Google KCC
- GitOps
- JWT Authentication

Lead AI and ML engineer and data scientist with **10+ years** of experience, building production AI/ML/LLM systems. Leads end-to-end AI delivery, combining hands-on model development with technical leadership to turn business problems into scalable, high-throughput, low-latency production systems. Experience spans predictive modelling, forecasting, operations research, computer vision, and generative AI across retail, cybersecurity, manufacturing, insurance, and supply-chain optimization. Selected contributions include modelling on **billion-row security datasets**, supply-chain optimization generating **~\$2.5M** in savings, and self-hosted LLMs saving an estimated **~\$300K** annually while improving availability during peak demand.

WORK EXPERIENCE

- Lead Artificial Intelligence Engineer** Best Buy | Jul 2025 – Present
 - Engineered a configuration-driven **LLM fine-tuning framework** supporting distributed training, task-specific adapters, and automated evaluation. Standardized task configuration and model artifacts enabled multiple tasks to share a common serving deployment.
 - Deployed **self-hosted LLM serving** with vLLM on GKE and **estimated annual API-cost savings of ~\$300K**, keeping model inputs within company-managed infrastructure and reducing exposure to external API availability and behavior during peak demand.
 - Built **production fraud-detection systems** spanning multiple models and batch, streaming, and real-time components, including a feature pipeline handling **40,000 events/sec** at peak. Supported a **200 ms scoring SLA** and improved fraud-score coverage by **~20%** during holiday traffic.
 - Lead architecture and production delivery of reusable LLM infrastructure; partner with data scientists to deploy secure, monitored retail AI services.
- Lead ML Engineer — ML Engineering & Operations** Wipro | Aug 2023 – Jul 2025
 - Engineered **AWS account digital fingerprinting** on NVIDIA DGX infrastructure using GPU-accelerated data processing. Trained per-entity behavioral autoencoders on **billion-row security datasets** to detect anomalies, with continuous retraining and model management.
 - Scaled **Proofpoint phishing-detection pipelines** to **2,400 emails/sec**, double the previous rate. Incorporated analyst feedback and Microsoft Sentinel integration to identify phishing missed by initial filters and support quarantine response.
 - Automated **abuse-inbox triage** with ML routing and OCR extraction across email bodies and attachments. Routed scammer identifiers to security teams and mail partners for takedowns, **doubling the prior case volume** and reducing manual review.
 - Led and mentored ML/MLOps engineers; coordinated client/NVIDIA delivery; screened **300+ resumes** and conducted **50+ technical interviews**. Received Best Buy AML Hall of Fame recognition (2024) during the Wipro client engagement.

MONITORING, TESTING & REPORTING

- Prometheus
- Grafana
- Google Cloud Monitoring
- Model Monitoring
- Load Testing
- Unit / Integration Testing
- Streamlit
- Power BI

LANGUAGES, DATA & STORAGE

- Python
- SQL
- Apache Beam / Dataflow
- Pub/Sub
- BigQuery
- MS SQL Server
- Redis Caching
- GCS
- Azure Blob Storage
- Artifact Registry
- Pandas / NumPy
- Dask
- Git
- Linux / Shell Scripting

CERTIFICATIONS

- Google Cloud Certified
- Generative AI Leader
- Oracle Cloud Infrastructure 2024
- Generative AI Certified Professional
- Microsoft Certified
- Azure AI Engineer Associate
- Microsoft Certified
- Azure Data Scientist Associate
- Microsoft Certified
- Power BI Data Analyst Associate

EDUCATION

- B.E. Electrical & Electronics
- Andhra University, Visakhapatnam
- 2012 – 2016 · 80%

HONORS & AWARDS

- Best Buy — Talent Quotient 2026
- Fraud-score coverage +20% · ~\$300K estimated annual API-cost savings
- Best Buy client - AML Hall of Fame 2024 (Wipro engagement)
- Above & Beyond recognition
- Wipro — 6 further awards
- Above & Beyond · Pinnacle · Being Responsive x3 (2024–25)
- Accenture
- Delivery Excellence · Kudos x2
- Infosys
- Insta Award x2

DOMAIN & CLIENT BREADTH

- Retail & e-commerce · Cybersecurity · Supply chain & logistics · Semiconductor manufacturing · Market intelligence (15+ verticals) · Insurance.
- Clients across the US, Norway, South Korea, and the UK.

LEADERSHIP & DELIVERY

- ML system architecture · Team leadership & mentoring · Technical hiring · Cross-functional delivery · Stakeholder management.

- Senior Data Scientist — Data Science / Azure ML

Accenture | May 2021 – Sep 2023

Developed **product-quality prediction for BamaGruppen** on Azure ML, with automated retraining and stakeholder alerts. Increased truck inspections from **130 to 160/day**, defect detection by **17%**, and customer approval by **20%**, delivering **\$2.5M annual savings**.

Designed **last-mile vehicle routing** with OR-Tools under capacity, delivery-window, waiting-time, and store-demand constraints. Compared optimized routes with manual plans to improve resource utilization and reduce logistics costs.

Implemented **transport optimization** using mixed-integer assignment models and APIs, automating daily dispatch and replacing manual Excel planning.

Modelled **supply-chain and distribution-hub scenarios** across cost, freshness, carbon footprint, and warehouse volumes to support network-design decisions.

Led cross-functional delivery, mentored data scientists, and presented recommendations to Norwegian stakeholders.
- Data Scientist

GD Research Centre Pvt Ltd | May 2020 – May 2021

Developed domain-aware **GlobalData forecasts** and scenario analysis across **15+ industry verticals**, informing market-intelligence reports and resource-allocation decisions.
- Deep Learning Engineer

Tata Consultancy Services | Jul 2019 – Mar 2020

Built and deployed an **end-to-end ML system for semiconductor defect detection**, using distributed training on **~1 million images in ~3 hours** and achieving **90% classification accuracy**. Deployed the system at Amkor in South Korea and trained the client team to operate it.
- Senior Software Engineer — ML

Infosys | May 2016 – Jul 2019

Implemented **opportunity win/lose prediction** with logistic-regression scoring and data pipelines, integrating predictions into business dashboards for sales prioritization.

Delivered tested **insurance ID-card generation** workflows, from impact analysis and design through implementation, support, and team knowledge sharing.

OPEN-SOURCE PROJECT

vLLM Model Chooser: Built an interactive GPU-to-model sizing tool with VRAM, quantization, and KV-cache fit filters. [Live project.](#)